TEST REPORT

EV Battery Monitoring System Using SocketCAN

Problem Statement 2 — Design and Implement a Distributed System Using SocketCAN

Technology: Python, SocketCAN, Virtual CAN (`vcan0`)
Number of ECUs: 4

1. Test Objective

The objective of testing was to verify the communication, functional behavior, diagnostic functions, fault handling, and recovery of the distributed EV Battery Monitoring System. Testing was performed using four independent software ECUs communicating through the Linux virtual CAN interface `vcan0`.

2. Test Environment

The test environment consisted of:

- Battery Sensor ECU
- BMS Controller ECU
- Power Control ECU
- Diagnostic ECU
- Virtual CAN interface: `vcan0`
- CAN monitoring tool: `candump`

3. Functional Test Cases

Test ID	Test Case	Expected Result	Result
FT-01	Battery Sensor ECU operation	Sensor values are generated and CAN messages are transmitted.	PASS
FT-02	BMS Controller operation	Sensor data is received, decoded, and processed.	PASS

FT-03	Power Control operation	Control commands are received and actuator states are updated.	PASS
FT-04	Diagnostic ECU operation	CAN traffic and ECU status are monitored.	PASS
FT-05	CAN communication	Configured CAN messages are visible on `vcan0`.	PASS
FT-06	HMI operation	Battery and system status are displayed.	PASS

4. CAN Communication Test

The complete system was started and CAN traffic was monitored using `candump vcan0`. The final implementation contains 10 unique CAN message IDs.

CAN ID	Purpose	Result
0x100	Cell Voltage 1–3	PASS
0x101	Cell 4, Temperature and Current	PASS
0x102	Battery Pack Status	PASS
0x103	Sensor Health	PASS
0x200	Power Control Command	PASS
0x201	Charging Command	PASS
0x202	Cooling Command	PASS
0x300	Power Control Status	PASS
0x301	Contactor Status	PASS
0x400	Diagnostic Status	PASS

All configured CAN messages were observed during system operation.

5. Fault Test Cases

5.1 Missing Message Detection

Objective: To verify that the Diagnostic ECU detects loss of communication from an ECU.

Procedure: The Battery Sensor ECU was stopped while the remaining ECUs continued running. CAN traffic and the Diagnostic ECU output were observed.

Expected Result: Sensor messages should stop and the Diagnostic ECU should report a Sensor ECU timeout.

Actual Result: The Diagnostic ECU detected the missing sensor communication and generated a timeout fault.

Result: PASS

5.2 Sensor Fault Detection

Objective: To verify detection of an abnormal battery sensor value.

Procedure: An abnormal battery temperature was introduced into the sensor data.

Expected Result: The BMS Controller and Diagnostic ECU should detect the abnormal temperature. The controller should enter SAFE MODE.

Actual Result: The abnormal temperature condition was detected and the system responded using the defined safe-state behavior.

Result: PASS

5.3 ECU Node Failure

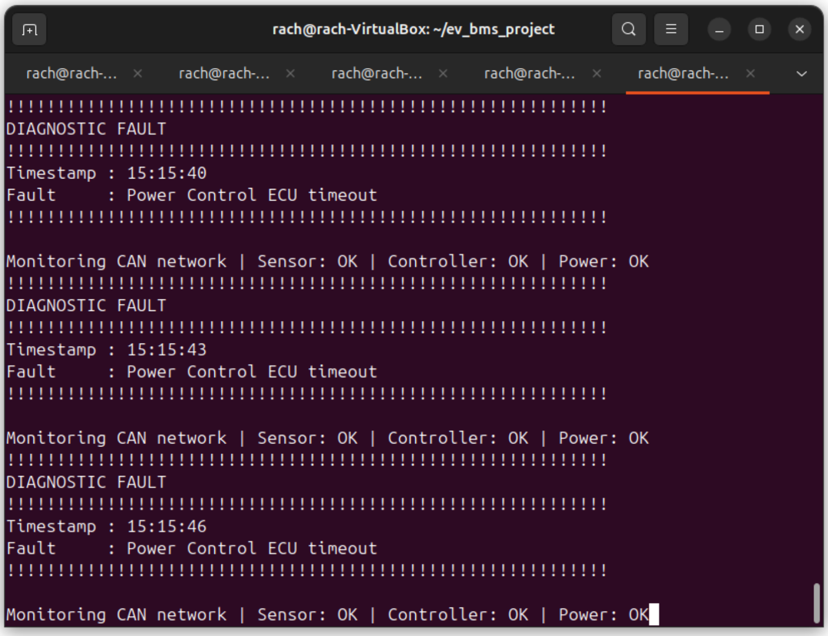
Objective: To verify detection of a failed Power Control ECU.

Procedure: The Power Control ECU was stopped during normal operation.

Expected Result: Power Control ECU status messages should stop and the Diagnostic ECU should report a Power Control ECU timeout.

Actual Result: The node failure was detected by the Diagnostic ECU while the remaining ECUs continued operating.

Result: PASS



5.4 ECU Recovery

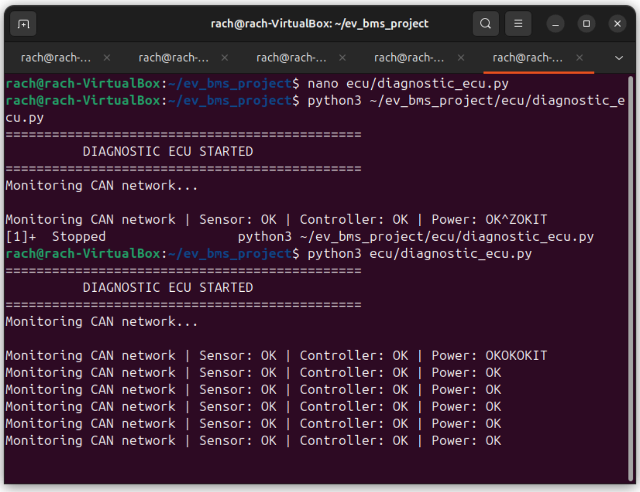
Objective: To verify recovery after an ECU failure.

Procedure: After the Power Control ECU failure was detected, the ECU application was restarted.

Expected Result: CAN communication should resume and the system should return to normal operation.

Actual Result: The Power Control ECU resumed CAN communication and normal system operation was restored.

Result: PASS



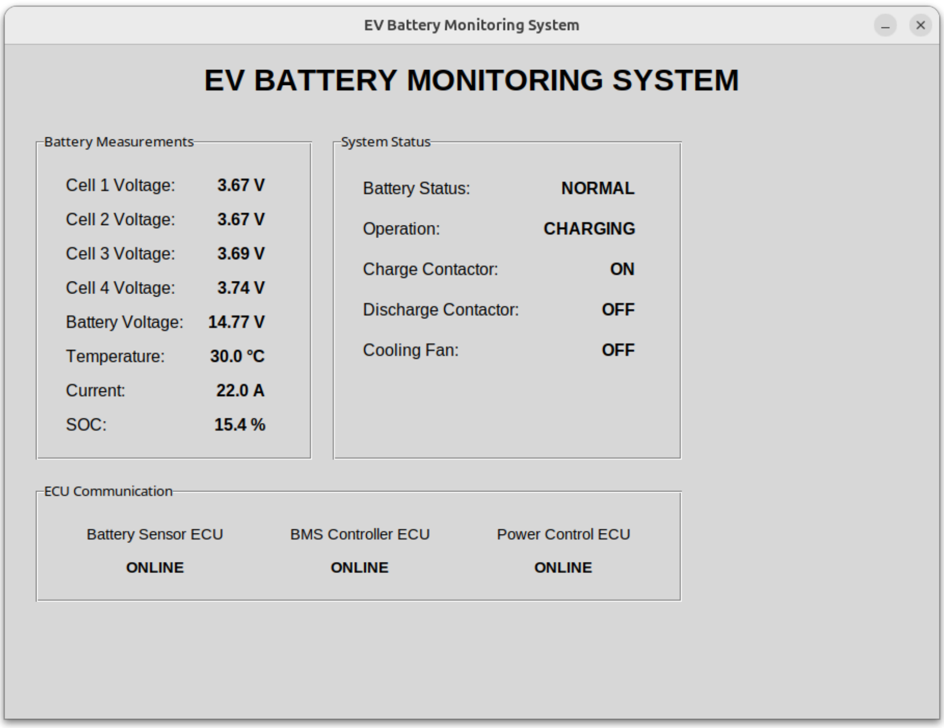
6. Integration Test

The complete system was tested with all four ECUs running simultaneously. Battery measurements were generated by the Battery Sensor ECU, processed by the BMS Controller, converted into control commands, and applied by the Power Control ECU. At the same time, the Diagnostic ECU monitored CAN communication and system health.

Expected Result: All ECUs should communicate correctly and the complete end-to-end system should operate as intended.

Actual Result: End-to-end communication and interaction between the four ECUs were successfully verified.

Result: PASS



7. Test Summary

Test Category	Tests Performed	Result
Functional Testing	6	PASS
CAN Communication	10 CAN IDs	PASS
Missing Message Detection	1	PASS

Sensor Fault Detection	1	PASS
ECU Node Failure	1	PASS
ECU Recovery	1	PASS
Integration Testing	1	PASS

8. Observations

- All four ECUs communicated successfully through the `vcan0` virtual CAN network.
- Ten unique CAN message IDs were implemented and observed.
- Periodic sensor, actuator-status, and diagnostic messages were transmitted successfully.
- Control messages were generated according to battery conditions.
- Missing ECU communication was detected by the Diagnostic ECU.
- Abnormal sensor conditions were detected and handled using SAFE MODE.
- The system detected Power Control ECU failure.
- The failed ECU successfully resumed communication after restart.

9. Conclusion

Testing confirmed that the EV Battery Monitoring System satisfies the required communication, functional, diagnostic, fault-handling, and recovery requirements.

The four independent ECUs successfully communicated through SocketCAN, and the system was validated under normal operation, missing-message, sensor-fault, node-failure, and recovery conditions.

The overall test result is PASS.